

Active Directory Home Lab

User Management & Domain Integration

Platform: Oracle VirtualBox | OS: Windows Server 2022 + Windows 11

Overview

This is a continuation of the Active Directory Home Lab setup. With the Domain Controller ready and configured, this section covers day-to-day tasks that IT helpdesk and sysadmin staff handle in a real-world work environment – things like creating user accounts, organizing them into groups, and connecting client machines to the domain.

These are some of the most common tasks in entry-level IT roles. Being able to demonstrate these hands-on in a home lab is a strong point for helpdesk, IT support, and junior sysadmin roles.

What This Covers

- Creating Organization Units (OUs)
- Adding user accounts and groups in Active Directory
- Joining a Windows 11 client machine to the domain
- Logging in as a domain user on the client machine

Part 1: Setting Up Organizational Units

Organizational Units (OUs) are folders in Active Directory used to organize users, computers, and groups. Keeping things separated into OUs makes it easier to apply Group Policies to specific departments or teams without affecting everyone else.

1. On the Windows Server, open the Start menu and go to **Windows Administrative Tools > Active Directory Users and Computers**.
2. In the left panel, expand and right-click on **LAB.local**, then select **New > Organizational Unit**.
3. Name the first OU **Accounts** and click OK. Leave **Protect container from accidental deletion** checked – this prevents the OU from being deleted by mistake.
4. Repeat the same steps to create a second OU named **Groups**.

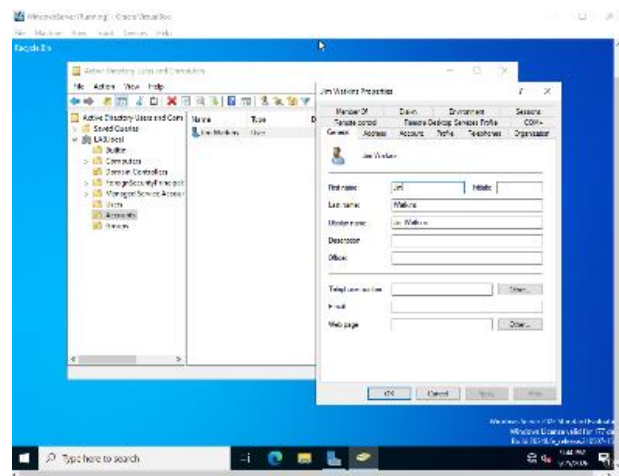
Note: Keeping users and groups in separate OUs is important because Group Policies can be applied per OU. This means you can set different rules for different departments without affecting the whole domain.

Part 2: Creating User Accounts

Adding a User

User accounts are how individuals get access to the domain. Every person who logs into a domain computer typically has an AD user account tied to them.

5. Right-click on the **Accounts** OU and select **New > User**.
6. Fill in the user's first and last name – for this example, **Jim Watkins**. Set the logon name to **jwatkins** (this is what the user types when logging in).
7. Click Next, enter a temporary password, and make sure **User must change password at next logon** is checked.
8. Click Next and then Finish. Jim Watkins will now appear under the Accounts OU.
9. To verify, double-click on **Jim Watkins** and check the **Member Of** tab to confirm if he is part of the **Domain Users** by default.



Note: Always leave “User must change password at next logon” checked. Admins should never know a user's personal password – that's a security violation. The temporary password is just to get them in the first time.

Adding Groups

Groups are used to assign permissions to multiple users at once instead of setting them individually. For example, everyone in the HR department gets access to HR files.

10. Right-click on the **Groups** OU and select **New > Group**.
11. Create groups as needed. For this lab, two groups were created: **IT** and **HR**.

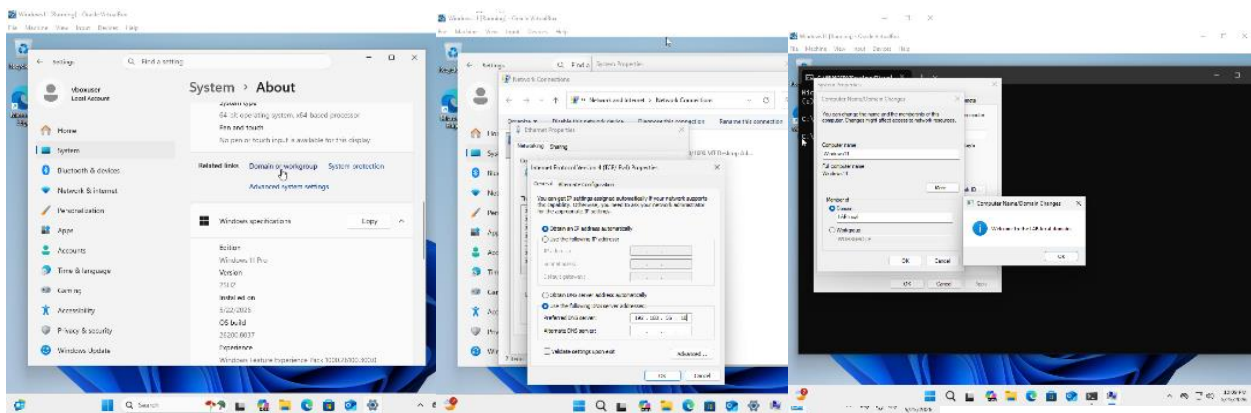
Part 3: Joining Windows 11 to the Domain

This is where the two VMs actually connect. Joining the Windows 11 machine to the domain means users created in Active Directory can now log into it using their domain credentials.

Point the Client DNS to the Server

Before joining the domain, the Windows 11 machine needs to be able to find the Domain Controller. This is done by pointing its DNS to the server's IP address.

12. On the Windows Server, open Command Prompt and run **ipconfig** to confirm the server's IP address (**192.168.56.10**).
13. On the Windows 11 VM, open Command Prompt and type **ncpa.cpl** to open Network Connections.
14. Right-click the **Ethernet Adapter > Properties > Internet Protocol Version 4 (TCP/IPv4) > Properties**.
15. Leave the IP address section on automatic (DHCP), but under the DNS section, manually enter the server's IP address (**192.168.56.10**) as the Preferred DNS Server.
16. Click OK to save.



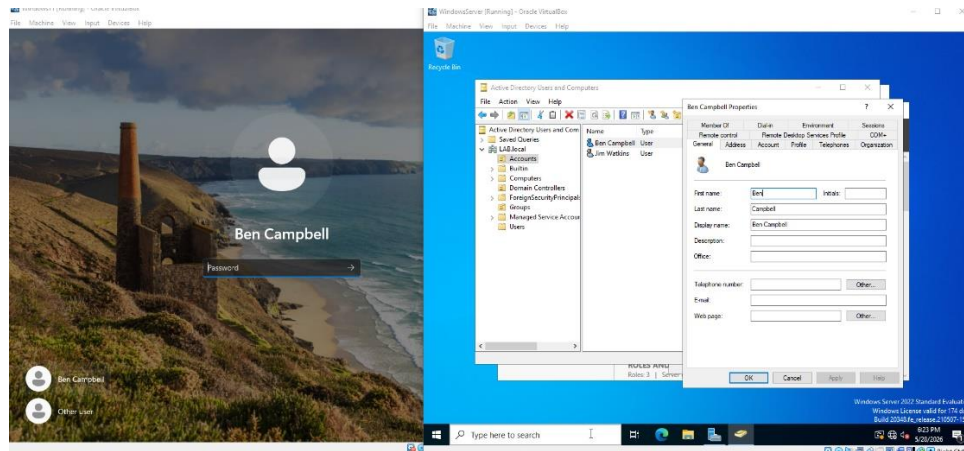
Note: Some settings in VirtualBox VMs may not show up through the normal Windows UI. Using Command Prompt commands like `ncpa.cpl` is a reliable way to access them directly.

Join the Domain

17. Open the Command Prompt and type `sysdm.cpl` to open System Properties.
18. Under the **Computer Name** tab, click **Change**. The machine will currently show as **Workgroup**.
19. Select **Domain** and type **LAB.local**, then click OK.
20. When prompted, enter the domain admin credentials to authorize the join.
21. Restart the Windows 11 machine to apply changes.

22. Back on the Server, open Active Directory Users and Computers and check under **LAB.local > Computers**. The Windows 11 machine should now appear there. If it doesn't show up immediately, right-click and hit Refresh.

Part 4: Logging In as a Domain User



With the client joined to the domain, any user account created in Active Directory can now log into the Windows 11 machine using their domain credentials.

23. On the Windows 11 login screen, select **Other User**.
24. Log in using the domain user's credentials – for example, username **BCampbell** and the temporary password set earlier.
25. Since **the user must change their password at the next login**, the user will be prompted to set a new password before getting in. This is expected behavior.

Key Takeaways

- OUs keep your AD environment organized and make Group Policy management easier down the line.
- Admins should never know a user's password – temporary passwords with forced reset on first login are the standard practice.
- The client machine's DNS must point to the Domain Controller before it can join the domain – without this, it can't find the domain at all.
- Command Prompt commands like `sysdm.cpl` and `ncpa.cpl` are useful workarounds when certain settings don't appear through normal Windows UI, especially in virtual environments.
- Once a machine is joined to the domain, any valid AD user account can log into it – not just a local account.